# The Birthday Paradox: Estimating Shared-Birthday Probabilities with Monte Carlo

VLADISLAV STEFANOV · FONTYS APPLIED IT · 24.10.2026

**Key result**

In a group of **23 people**, a shared birthday is more likely than not: about **51%** in the model.

## 1. Research question

**What is the probability that at least two people in a group of *n* people share the same birthday?**

## 2. System description

The system is a real group of people, such as a class, a team or a party. Each person has one birthday. The study examines how likely it is that any two of them share one as the group grows.

## 3. Model: real system → simplification

- Births vary by season → all 365 dates equally likely.
- February 29 is rare → it is excluded.
- Birthdays are unrelated → independent variables.
- Shared birthday (match) → same day and month.

## 4. Proof of concept: Monte Carlo

To solve the problem, **many random groups are simulated** in Python and the share of groups with a match is counted.

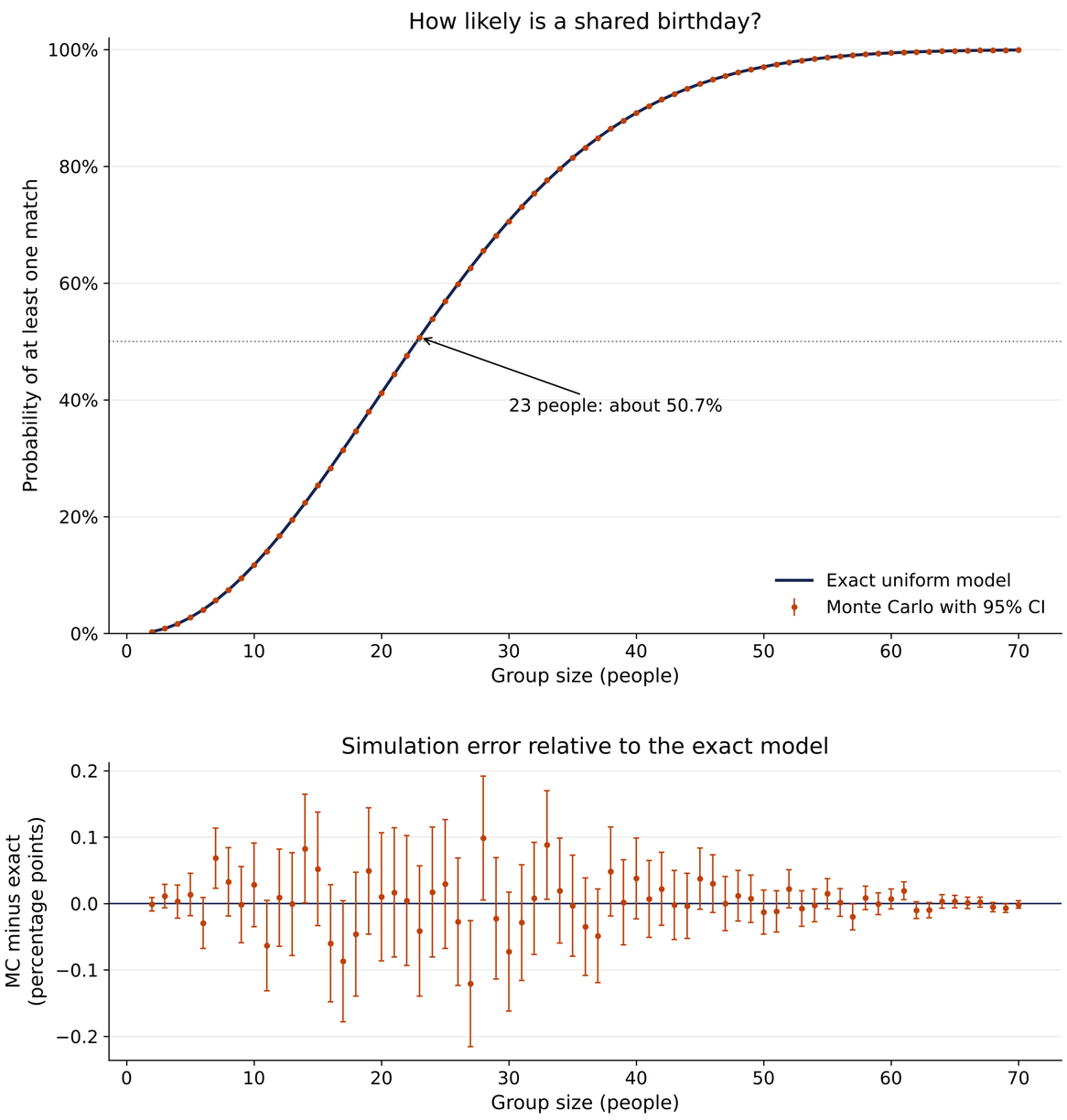
1. **Sample** *n* birthdays from 0 to 364 for one group
2. **Sort** the group so equal dates sit next to each other
3. **Compare neighbours**: two equal dates mean a match
4. **Repeat** for M = 1,000,000 groups per size, n = 2 to 70
5. **Estimate** p̂ = matching groups / M, plus its uncertainty

**Illustration of steps 2 and 3 for a group of 6 (example values, not simulation output)**

| 12 | 45 | 45 | 130 | 201 | 340 |
|---|---|---|---|---|---|

Neighbours 45 = 45, so this group is recorded as **True** (one match).

## 5. Results: probability by group size




**23**
smallest group where a match is more likely than not

**97.0%**
chance of a match in a group of 50 (simulated 97.024%)

**69**
group sizes simulated (2 to 70), each with 1,000,000 groups

**61 / 69**
95% intervals that contain the exact probability

**Figure 1.** Birthday-match probabilities for group sizes 2 to 70 using 1,000,000 simulated groups per size. Upper panel: exact probabilities and Monte Carlo estimates with approximate 95% confidence intervals (too narrow to see at this scale). Lower panel: signed estimation errors in percentage points (pp) with the same uncertainty. Intervals crossing zero contain the exact probability.

## 6. Precision of the estimate

Each simulated group is a yes/no outcome, so the estimate has a binomial standard error (SE). For M groups:

$$SE = \sqrt{\hat{p}\,(1 - \hat{p}) / M}$$

$$95\%\ \text{confidence interval (CI)} \approx \hat{p} \pm 1.96 \cdot SE$$

Normal approximation is valid at every size (≥ 10 matching and non-matching groups). Intervals are per group size.

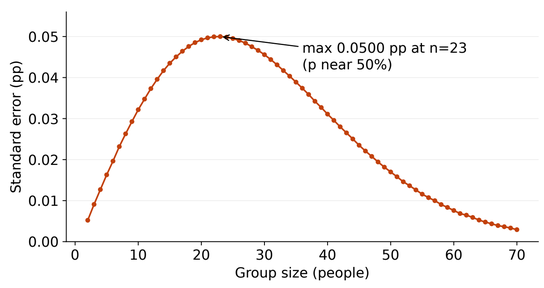



**Figure 2.** SE is largest where p̂ is near 50% (n = 3).

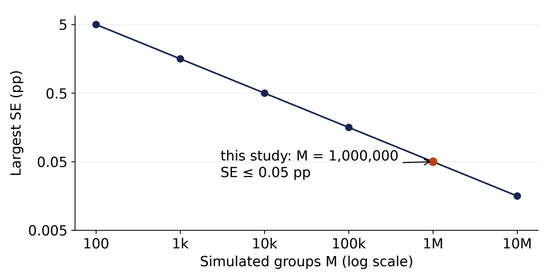



**Figure 3.** Largest possible SE (p = 0.5) from the formula.

**Takeaway:** precision grows with √M. Halving the error needs 4× more runs; 1,000,000 runs give SE ≤ 0.05 pp.

## 7. Selected results

| n | MC estimate | Approx. 95% CI | Exact |
|---|---|---|---|
| 10 | 11.723% | 11.660 – 11.786% | 11.695% |
| 22 | 47.574% | 47.476 – 47.672% | 47.570% |
| **23** | **50.688%** | **50.590 – 50.786%** | **50.730%** |
| 30 | 70.559% | 70.470 – 70.649% | 70.632% |
| 50 | 97.024% | 96.991 – 97.058% | 97.037% |
| 70 | 99.914% | 99.909 – 99.920% | 99.916% |

Values from birthday_results.csv; M = 1,000,000 groups per size.

## 8. Interpretation and limitations

Real birthdays are not spread evenly. Theory shows that unequal frequencies can only raise the chance of a match, so these model values are likely a **lower bound** for real groups.

### Why only 23 people?

The question is not "does someone share *my* birthday?" but "does **any pair** share one?". A group of 23 already has 253 pairs.

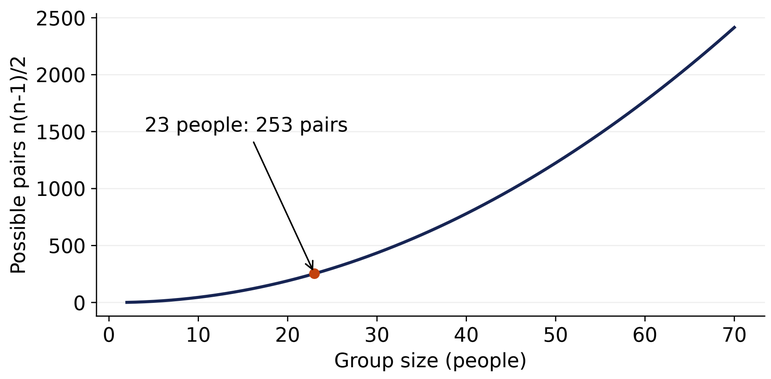



**Figure 4.** Possible pairs, n(n − 1)/2, grow with n², not n.

### Limitations

- Uniform, independent birthdays are assumptions; no real birth data were analysed.
- Intervals measure simulation error only, not error in the assumptions.
- A fixed seed makes the run reproducible but does not remove sampling error.

## 9. Verification

This model also has an exact solution, which is used to check the code. It counts groups in which everyone has a different birthday:

$$P(\text{no match}) = \prod_{k=0}^{n-1} (365 - k) / 365$$

$$P(\text{match}) = 1 - P(\text{no match})$$

The exact value lies inside 61 of 69 separate 95% intervals; the largest difference is 0.121 pp. Occasional misses are expected.

## 10. Real-world use cases

The birthday problem describes any **collision**: many items assigned at random to a fixed number of categories.

- **Hash functions:** a collision becomes likely after about $2^{b/2}$ inputs for a b-bit hash (the "birthday attack"), so hashes must be long.
- **Random IDs** (e.g. UUIDs, voucher codes): the ID space must be large enough that duplicates stay unlikely.
- **Record matching:** in a large database, some pair of records often agrees by chance, even when a match between two specific records is rare.

## 11. Conclusion

In a group of 23 people a shared birthday is already more likely than not; in a class of 30 it is about 7 in 10. The Monte Carlo PoC matches the exact model to within 0.121 pp. Because real birthdays are not spread evenly, the chance in real groups is likely slightly higher than these model values. **Next step:** use real birth-frequency data (only the sampling step changes).

## 12. References

1. University of Chicago. *Introduction to Data Science: Probability definitions and rules*. ds1.datascience.uchicago.edu/10/1/Probability_1_RulesDefinitions.html
2. NumPy. *Random sampling* documentation. numpy.org/doc/stable/reference/random/
3. Menezes, A., van Oorschot, P. and Vanstone, S. (1996). *Handbook of Applied Cryptography*, §2.1.5 Birthday attacks. CRC Press.

### Reproducibility

Python 3.14.6, NumPy 2.5.4, matplotlib 3.11.2; PCG64, seed 1. Figures 1 and 2 from the results file (Figure 1 re-plotted with x-axis labels on both panels); Figures 3 and 4 from formulas.